Topic 1 — Environment (पर्यावरण) Basics

Covers syllabus (click to expand)

Environment – Meaning | Environmental Components | Classification of Environment | Human–Environment Interaction

Sources baked in: NCERT Class 11–12 (Biology Ch 13–16, Geography Ch 14–15), MoEFCC, CPCB, UNEP, Brundtland Report, UPPCS Prelims PYQs 2018–2025 **Weight:** ★★★ High — Prelims direct + Mains foundation **Last verified:** August 2026

Current Affairs

Year	Fact	Why asked
2023	**Amrit Dharohar Scheme** — wetlands conservation; launched on WED 2023	Wetlands + governance
2023	**National Green Hydrogen Mission**	SD / climate policy overlap
2019	**NCAP** (राष्ट्रीय स्वच्छ वायु कार्यक्रम) — 132 cities; 20–30% PM2.5/PM10 reduction target by 2024	Urban (नगरीय) environment governance

1.1 Environment – Meaning

Scope: NCERT definition | four spheres | atmosphere basics | scales & types | milestones & law

Source	Definition
NCERT (Class 12)	Sum of all **biotic + abiotic** factors influencing living organisms ← **most asked**
UNEP	Physical, chemical, and biotic factors acting on organism/community, determining form and survival
EPA 1986 (पर्यावरण संरक्षण 1986) (Legal)	Water, air, land + inter-relationship among them and with humans, other living beings, plants, and property
General	External conditions — physical, chemical, biological, social, economic, cultural — affecting life

- French *environ* means surroundings (परिवेश).
- In UPPCS, "environment" in the scientific sense means **natural/physical** factors (biotic plus abiotic). Social and economic factors are not the primary focus in most Prelims questions.
- The **lithosphere** is Earth's solid land crust. It covers about **29%** of the surface.

- The **hydrosphere** is all water on Earth in liquid, ice, and vapour form. It covers about **71%** of the surface.
- The **atmosphere** is the gaseous envelope. Roughly **78%** is nitrogen and **21%** is oxygen.
- The **biosphere** is the zone where life exists. It spans from about **200 m below soil** to about **6 km above sea level**, where lithosphere, hydrosphere, and atmosphere meet.
- **Pedosphere** (soil) is usually grouped with the lithosphere or with edaphic abiotic factors. It is not counted as a fifth sphere.
- The **troposphere** (क्षोभमंडल) extends from the surface to about **12 km**. All weather phenomena occur here.
- The **stratosphere** (समतापमंडल) lies roughly **12–50 km** above the surface. The **ozone layer** that absorbs UV radiation sits in this layer.
- **Micro-scale** environment covers immediate surroundings such as a room or a pond edge.
- **Meso-scale** covers a village or city.
- **Macro-scale** covers regional or global systems.
- **Total environment** can be natural, social, economic, or cultural. Competitive papers usually default to **natural/physical** environment.
- The **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन 1972) was the first major UN conference on the human environment.
- **UNEP** (United Nations Environment Programme) was created at Stockholm. Its headquarters is in **Nairobi, Kenya**.
- **World Environment Day (WED)** is observed every year on **5 June**. The date marks the opening day of the Stockholm Conference in 1972.
- **MoEF** was established in **1985**. It was renamed **MoEFCC in 2014** when climate change (जलवायु परिवर्तन) was added to its mandate.
- The **Environment (Protection) Act, 1986** is India's umbrella environmental law. Parliament passed it after the **Bhopal Gas Tragedy (1984)**.
- **Article 48A** is a Directive Principle. The State shall protect the environment and safeguard forests and wildlife. The **42nd Amendment (1976)** added it.
- **Article 51A(g)** is a Fundamental Duty. Every citizen shall protect and improve the natural environment. The same amendment added it.
- **Article 21** guarantees the Right to Life. The Supreme Court expanded it to include the **right to a wholesome and healthy environment** in cases such as *M.C. Mehta* and *Subhash Kumar*.

> **Logic:** UPPCS 2018 treats environment as **जैविक + अजैविक**. Trap options say "only physical" or "only living." **Article 48** deals with agriculture and cows. Environment is **Article 48A**.

PYQ — UPPCS Prelims 2025, Q62

Which of the following pairs is/are NOT correctly matched?

(Day) — (Date)

1. International Ozone Day — 16 September
2. World Environment Day — 5 June

3. World Wildlife Day — 22 March

A. 1 and 2

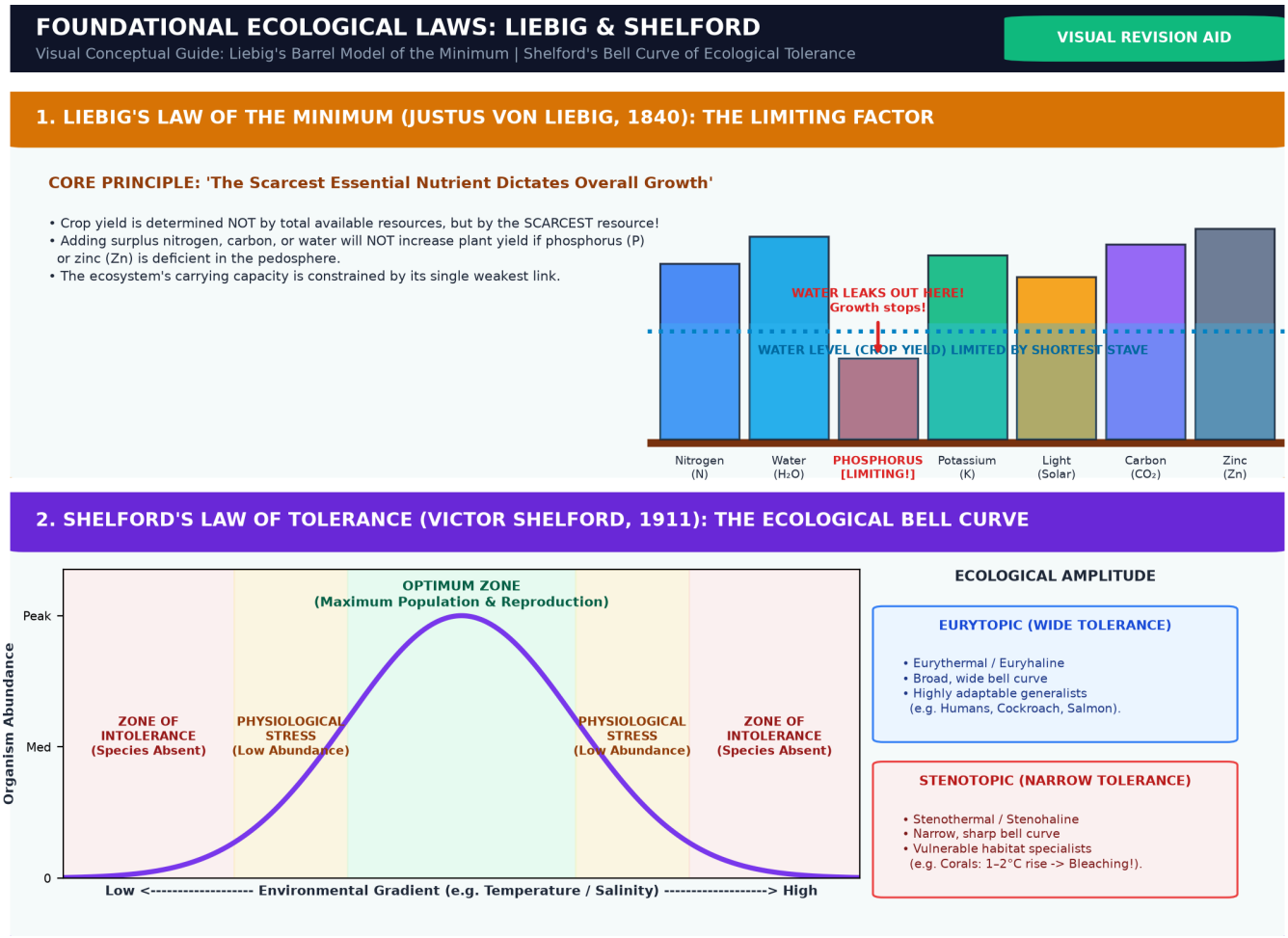
B. Only 3

C. 2 and 3

D. Only 1

Show answer

Ans: B — Only pair 3 is wrong. World Wildlife Day = **3 March**, not 22 March.



1.2 Environmental Components

Identity: Environment = Abiotic + Biotic | Liebig + Shelford | N-fixation

Abiotic factor category	Includes	Hindi
Edaphic	Soil texture, minerals, pH, moisture	मृदा संबंधी
Climatic	Temperature, rainfall, wind, humidity	जलवायु संबंधी

Abiotic factor category	Includes	Hindi
Topographic	Altitude, slope, aspect	स्थलाकृति संबंधी
Biotic group	Also called	Function
---	---	---
Producers	Autotrophs	Convert solar energy → chemical energy
Consumers	Heterotrophs	Feed on other organisms
Decomposers	Saprotrophs	Break down dead matter; recycle nutrients
Detritivores	—	Feed on detritus (earthworms, termites)

- About **99%** of biosphere energy comes from the **Sun**.
- **Liebig's Law of Minimum** states that growth is controlled by the **scarcest essential resource**. That resource is the limiting factor.
- **Shelford's Law of Tolerance** (शेलफोर्ड का सहनशीलता नियम) states that each species survives only within a **minimum–maximum range** for each factor. Beyond that range, stress or death follows even if other factors are abundant.
- **N_2** makes up about **78%** of air, but most life cannot use it directly. Organisms need **nitrogen fixation** first.
- **Rhizobium** bacteria live in legume root nodules and fix atmospheric N_2. Pulses, gram, and soybean are common legume examples.
- **Azotobacter** is a free-living soil bacterium that fixes nitrogen.
- **Cyanobacteria** (blue-green algae) also fix nitrogen in soil and water.
- **Humus** is organic matter in soil. It comes from decomposed plant and animal material, so it is **not** an abiotic component.
- Abiotic factors shape biotic life. Desert heat favours cacti and other drought-adapted plants.
- Biotic activity modifies abiotic conditions. Forests increase local rainfall. Overgrazing can trigger desertification.
- **Dissolved Oxygen (DO)** is a key aquatic abiotic indicator. Fish die in polluted rivers when DO falls too low.

> **Logic:** Liebig names **which single factor limits growth**. Shelford names the **survivable range**. Humus is organic, not abiotic.

PYQ — UPPCS Prelims 2019, Q46

Nitrogen fixing bacteria make combination with cells of the roots of

A. Pulses

B. Rice

C. Wheat

D. Sugarcane

Show answer

Ans: A — Rhizobium associates with leguminous (pulse) root nodules.

1.3 Classification of Environment

Identity: Physical/Biological/Cultural triad | Natural/Anthropogenic | Terrestrial/Aquatic | Internal/External | India = 10 biogeographic zones

Basis	Categories
Standard triad	**Physical** / **Biological** / **Cultural** (**Operational** is **not** a category)
Origin	Natural vs **Anthropogenic** (human-modified — cities, dams, cropland)
Components	Abiotic vs Biotic
Domain	**Terrestrial** (~29% surface) vs **Aquatic** (~71%)
Organism level	**Internal** (inside body — physiology) vs **External** (outside — ecology default)
Scale	Micro / Meso / Macro
Human use	Rural / Urban / Industrial

Physical, biological, and cultural environment

- In the standard triad, environment is classified as **physical**, **biological**, and **cultural**.
- The **physical environment** (भौतिक पर्यावरण) includes landforms, water bodies, climate, soils, rocks, and minerals.
- The **biological environment** (जैविक पर्यावरण) includes plants, animals, and microorganisms.
- The **cultural environment** includes man-made features such as economic, social, and political elements.
- **Operational environment** is **not** part of this standard classification.
- **Internal environment** (आंतरिक पर्यावरण) means conditions inside an organism's body, such as blood chemistry and tissue fluids studied in physiology.
- **External environment** (बाह्य पर्यावरण) means conditions outside the organism. Ecology usually studies the external environment.

Aquatic type	Description	Indian example
Marine	Oceans/seas — high salinity	Arabian Sea, Bay of Bengal
Lentic	**Standing** freshwater	Loktak Lake (लोकटक), Dal Lake
Lotic	**Flowing** freshwater	Ganga, Yamuna
Estuarine	River meets sea — brackish	Sundarbans (सुंदरबन) delta (डेल्टा)

Aquatic type	Description	Indian example
Wetland	Transitional — neither fully land nor water	Chilika, Keoladeo (केवलादेव)
Marine zone	Location	Key fact
----------	----------	----------
Neritic	Shallow shelf (0–200 m)	Fisheries, coral, mangroves
Pelagic	Open water column	Plankton, tuna
Benthic	Sea floor	Crabs, decomposers
Littoral	Intertidal shore	Barnacles, molluscs

Transitional ecosystems — mangroves & wetlands

- A **wetland** is land that stays waterlogged or seasonally flooded, so it is neither fully terrestrial nor fully aquatic.
- Wetlands store floodwater, recharge groundwater, filter pollutants, and support migratory birds.
- **World Wetlands Day** is observed on **2 February** (Ramsar (रामसर) Convention anniversary).
- India's wetland rules are notified under **Wetlands (Conservation and Management) Rules, 2017**.
- Classic Indian wetlands s include **Chilika** (Odisha) and **Keoladeo** (केवलादेव) (Rajasthan).
- A **mangrove** is a salt-tolerant coastal forest growing in intertidal / brackish zones.
- Mangroves act as a storm buffer, reduce shoreline erosion, and nursery fish stocks.
- India's largest mangrove stretch is the **Sundarbans** (सुंदरबन) (West Bengal / Bangladesh).
- In classification questions, mangroves are **coastal transitional**, not "purely terrestrial."
- UPSC (संघ लोक सेवा आयोग) 2020 trap: among Desert, Mangrove, Rainforest — only Desert and Rainforest count as terrestrial.

> **Logic:** Mangrove / wetland = transitional. Do not tick mangrove as terrestrial.

Urban heat island

- An **urban heat island** means a city runs warmer than nearby rural land under similar weather.
- Typical city–rural gap s is about **2–5°C**.
- Concrete and asphalt absorb heat and re-radiate it at night.
- Dense buildings trap warm air and block wind flow.
- Less vegetation cuts evapotranspiration cooling, so city nights stay warmer than rural nights.
- Extra heat raises energy demand for cooling and worsens ground-level air pollution stress.
- Delhi NCR is a frequent Indian example of urban heat + air-quality interaction.

> **Logic:** Urban heat island = city warmer by roughly **2–5°C**, not a separate climate zone.

India's 10 biogeographic zones

Authors / year: Rodgers & Panwar | **1988** (WII delineation used s)

#	Zone	feature
1	**Trans-Himalaya**	Cold desert — Ladakh (लद्दाख), high-altitude sparse vegetation
2	**Himalaya** (हिमालय)	Alpine to subtropical belts along the main Himalayan range
3	**Desert**	Thar (थार) — Rajasthan and adjoining Gujarat
4	**Semi-Arid**	Dry scrub/savanna (सवाना) fringe of Gujarat–Maharashtra–central India
5	**Western Ghats** (पश्चिमी घाट)	Biodiversity hotspot (जैव विविधता हॉटस्पॉट); high endemism
6	**Deccan (दक्कन) Peninsula**	Plateau forests and dry deciduous tracts
7	**Gangetic Plain**	Fertile alluvial (जलोढ़) plains — covers **most of UP**
8	**Coasts**	Mangroves, estuaries, coral reefs
9	**North-East India**	Indo-Malayan biodiversity corridor
10	**Islands**	Andaman (अंडमान) & Nicobar (निकोबार) and Lakshadweep (लक्षद्वीप)

- Remember: the number **10**, not 8 or 14. Fourteen is a global biome (जीवोम) count from a different classification list.
- Most of Uttar Pradesh falls in the **Gangetic Plain** zone.
- The northern Terai (तराई) fringe of UP links to Himalayan foothill ecology.

> **Logic:** UPPCS has asked "how many biogeographic zones in India?" → **10** (Rodgers & Panwar, 1988).

CRZ — Coastal Regulation Zone

Legal basis: Notifications under the **Environment (Protection) Act, 1986** (nodal: MoEFCC)

- CRZ regulates development along India's coast to protect ecology and coastal livelihoods.
- The coastal stretch covered is generally from the High Tide Line landward and seaward as defined in the notification (focus is categories, not metre-by-metre measurement).
- **CRZ-IA** covers ecologically sensitive areas such as mangroves, coral reefs, and nesting beaches — **no new development** of the usual commercial kind.
- **CRZ-IB** is the **intertidal zone** between Low Tide Line and High Tide Line.
- **CRZ-II** covers already **developed / urbanised** coastal stretches within municipal limits.
- **CRZ-III** covers **rural** coastal areas where development is more restricted than in CRZ-II.
- The **CRZ 2019** notification revised earlier CRZ rules. Papers still test the **IA / IB / II / III** category logic.

> **Logic:** CRZ-IA = most sensitive (mangroves and corals). CRZ-II = developed urban coast. CRZ-III = rural coastal belt.

Eco-Sensitive Zones (ESZ)

Legal basis: Declared under the **Environment (Protection) Act, 1986**

- An **ESZ** is a **buffer belt** around a National Park, Wildlife Sanctuary, or other protected area.
- Eco-Sensitive Zones are notified under the **Environment (Protection) Act, 1986**, **not** under the Wildlife (Protection) Act, 1972.
- The aim is to reduce shock between a strict protected core and surrounding human use.
- The purpose is **regulation**, not a total ban on every human activity except agriculture.
- **Permitted** activities typically include ongoing local agriculture/horticulture, rainwater harvesting, and organic farming.
- **Regulated** activities include tree felling, hotels/resorts, commercial use of natural water, and road widening.
- **Prohibited** activities include polluting discharges into natural water bodies and tourism such as flying over protected areas in an aircraft or hot-air balloon.
- ESZ width is **not** a fixed all-India kilometre figure. It is notified case by case, often up to about **10 km**.
- Do not treat 10 km as a universal law for every protected area.
- An ESZ is an **EPA buffer tool**. It is **not** the same as declaring a National Park under the Wildlife (Protection) Act.

> **Logic:** ESZ = EPA 1986 buffer. "All human activity banned except agriculture" is **false**. CRZ = coastal regulation under the same Act.
>
> **Logic:** Lentic = lake or pond (standing water). Lotic = river or stream (flowing water). marks **river** as lotic.

PYQ — UPPCS Prelims 2023, Q112

Which of the following is an example of a **Lotic Ecosystem** (प्रवाही जल)?

A. Swamps

B. Ponds

C. Rivers

D. Marshy areas

Show answer

Ans: C — Lotic = flowing water. Ponds/swamps = lentic or wetland.

1.4 Human–Environment Interaction

Identity: Reciprocal relationship | models | carrying capacity | SD governance | Indian movements

Model	Proponent	Idea	Status
Environmental Determinism	Ratzel	Environment **dictates** culture/society	Largely **rejected**
Possibilism	Vidal de la Blache	Environment offers **possibilities**; humans choose	**Accepted**
Neo-determinism	Griffith Taylor	Humans modify environment **within limits**	Balanced view
Concept	Definition / fact		
---	---		
Carrying capacity	Maximum load an environment can sustain indefinitely. Resource extraction must stay **below** regeneration. Waste must stay **within** absorption capacity		
IPAT	Impact = **P**opulation × **A**ffluence × **T**echnology		
Ecological footprint (पारिस्थितिक पदचिह्न)	Land/water area needed to support consumption — measured in **global hectares (gha)**		
Tragedy of the Commons	Shared resources overexploited when individual interest > collective good — **Garrett Hardin, 1968**		
Sustainable Development	"Meet present needs without compromising future generations' ability to meet theirs" — **Brundtland Report, 1987** (*Our Common Future*)		

Brundtland Report, 1987

Commission: World Commission on Environment and Development (WCED) | Chair: Gro Harlem Brundtland

- The report's official title is ***Our Common Future*** (published **1987**).
- It popularised the classic definition of **Sustainable Development**.
- SD balances three pillars: economic growth, social equity, and environmental protection.
- The commission worked roughly **1983–1987** under a UN mandate.
- Do **not** confuse Brundtland with ***The Limits to Growth*** (Club of Rome, **1972**).
- UPPCS 2024 A/R trap: Reason saying Brundtland = *Limits to Growth* is **false**.

> **Logic:** Brundtland 1987 = *Our Common Future*. Club of Rome 1972 = *Limits to Growth*.

Rio Earth Summit, 1992

Also called: United Nations Conference on Environment and Development (UNCED) | Place: Rio de Janeiro

- Rio shifted global environment talk from awareness (Stockholm 1972) to an action framework.
- Three landmark outcomes signed/adopted (अंगीकृत) at Rio are **UNFCCC**, **CBD**, and **Agenda 21**.
- **UNFCCC** is the climate convention that later produced Kyoto (क्योटो) and Paris pathways.
- **CBD** is the Convention on Biological Diversity for conservation and benefit-sharing.
- Rio is **1992**, not Stockholm 1972 and not Agenda 21's false "1995" year.

> **Logic:** Rio 1992 package = UNFCCC + CBD + Agenda 21.

Agenda 21, 1992

- **Agenda 21** is a non-binding action blueprint for sustainable development in the 21st century.
- It was adopted at the **Rio Earth Summit, 1992**.
- It covers local, national, and global action themes (poverty, resources, governance, participation).
- The title **Agenda 21** refers to the 21st century action plan. It does **not** mean the document was signed in the year 2021.
- UPPCS 2025 A/R used the wrong year **1995** as a Reason trap — that Reason is **false**.

> **Logic:** Agenda 21 = **1992** (Rio). Never mark 1995.

LiFE Mission, 2022

Full form: Lifestyle for Environment (जीवनशैली पर्यावरण) | Nodal idea: India at climate diplomacy

- PM Modi proposed the LiFE idea at **COP26, Glasgow (2021)**.
- India formally launched **Mission LiFE** in **June 2022**.
- The mission pushes individual and community lifestyle shifts — save energy, cut waste, choose sustainable food and consumption.
- Notes often cite about **75** pro-planet actions / behavioural pledges under LiFE.
- The Mission LiFE app / campaign tracks and popularises those behaviours.
- Trap: "proposed at COP-25 Madrid" is **false**.

> **Logic:** Idea = COP26 (2021). Launch = **2022**. Not COP-25 Madrid.

Chipko Movement (चिपको आंदोलन), 1973

Place: Reni village, Chamoli (चमोली) (then Uttar Pradesh; now **Uttarakhand** (उत्तराखंड))

- Chipko means villagers hugged trees to stop commercial felling.
- The 1973 Reni episode is the classic anchor for the movement's start.
- **Sunderlal Bahuguna** was a leading Chipko activist linked to the Uttarakhand forest protests.

- **Chandi Prasad Bhatt** (चंडी प्रसाद भट्ट) was another key Chipko leader from the Garhwal (गढ़वाल) region.
- Women played a visible role in forest protection protests.
- Chipko is often cited as **passive resistance** (निष्क्रिय प्रतिरोध) / non-violent environmental action.
- Place trap: Chipko is **not** Assam, Kerala, or Karnataka.

> **Logic:** Chipko = **1973**, Reni, Uttarakhand (then UP). Appiko (अप्पिको) = Karnataka.

Appiko Movement, 1983

Place: Karnataka (Western Ghats / Uttara Kannada belt)

- Appiko is the southern Indian analogue of Chipko.
- Villagers hugged trees to resist forest felling and protect local ecology.
- UPPCS 2025 asked the state association — correct fact is **Karnataka only**.
- Do not mark Uttarakhand, UP, or Kerala for Appiko.

> **Logic:** Appiko = **Karnataka, 1983**. Chipko = Uttarakhand.

Joint Forest Management (JFM) (संयुक्त वन प्रबंधन), 1990

- JFM is a partnership between the **Forest Department** and local **village communities**.
- The national policy push is dated to the **1990** Government of India resolution / circular.
- Village communities help protect and regenerate forests under JFM.
- In return, communities share usufruct and other forest benefits.
- JFM is participatory forest governance, not a full ownership transfer law.
- Do not confuse JFM (**1990**) with FRA (**2006**) or EPA (**1986**).

> **Logic:** JFM launch year fact = **1990**.

Forest Rights Act (FRA), 2006

Full title: Scheduled Tribes and Other Traditional Forest Dwellers (Recognition (मान्यता) of Forest Rights) Act, 2006

- FRA recognises **individual** and **community** forest rights of Scheduled Tribes and **OTFDs**.
- OTFD means Other Traditional Forest Dwellers who depended on forests for livelihood.
- Rights can include habitation, cultivation, and community forest resource use, subject to the Act's process.
- FRA is framed as justice for forest dwellers, not as a blanket free-for-all to clear forests.
- Year trap: FRA = **2006**, after JFM 1990 and EPA 1986.

> **Logic:** FRA 2006 = tribal (आदिवासी)/OTFD forest rights. JFM 1990 = community–State forest partnership.

- **Polluter Pays** and **Precautionary Principle** entered Indian environmental jurisprudence through *Vellore Citizens Welfare Forum* (**1996**).
- The **Public Trust Doctrine** treats air, water, and forests as held by government in trust for the public.
- The **Bhopal Gas Tragedy (1984)** triggered the umbrella **Environment (Protection) Act, 1986**.

Landmark Environmental Classics & Pioneer Thinkers

Thinker / Landmark	Year / Publication	Core Concept & Exam Significance
Rachel Carson	**1962** (*Silent Spring*)	Exposed environmental hazards of indiscriminate synthetic pesticides (**DDT** bioaccumulation and eggshell thinning); catalytic milestone that birthed the modern global environmental movement.
John W. Bennett	**1976** (*The Ecological Transition*)	Coined and developed the concept of **'Ecological Transition'**, analyzing human society's cultural shift from balanced ecological adaptation to intensive environmental exploitation.
Garrett Hardin	**1968** (*The Tragedy of the Commons*)	Postulated in *Science* that independent rational actors deplete shared public resources (pastures, fisheries, clean air) through unconstrained self-interest.
Club of Rome	**1972** (*The Limits to Growth*)	Commissioned MIT computer-simulation model (Meadows et al.) showing ecological collapse if exponential economic and population growth outpace planetary carrying capacity. *(Trap: Brundtland Report 1987 is Our Common Future, NOT Limits to Growth).*
Ernst Haeckel	**1866** (*Generelle Morphologie*)	Coined the term **'Ecology'** (*Oekologie*) for the study of organism-environment interactions.
A.G. Tansley	**1935**	Coined the term **'Ecosystem'** as the fundamental functional unit of nature.
Arne Naess	**1973**	Coined **'Deep Ecology'** (advocating inherent biocentric rights of all living species) versus utilitarian 'Shallow Ecology'.
Joseph Grinnell	**1917**	Coined and formalized the concept of **'Ecological Niche'** as the functional role and distributional unit of a species.
Mathis Wackernagel & William Rees	**1996**	Formulated the concept of **'Ecological Footprint'**, measuring human resource appropriation in global hectares (gha).

Ecological Zonation & Organism Habitat Adaptations

- **Organism Adaptations**:
- **Arboreal**: Tree-dwelling animals adapted for life in tree canopies (e.g., squirrels, chameleons, monkeys).
- **Fossorial**: Animals adapted for digging and living underground in burrows (e.g., moles, pocket gophers, earthworms).

- **Xericole**: Animals anatomically and physiologically adapted for arid, desert environments with minimal water loss (e.g., camels, kangaroo rats, desert lizards).
- **Cursorial**: Animals adapted specifically for running fast on land (e.g., cheetah, ostrich, deer).
- **Lakes & Freshwater Zonation**:
- **Littoral Zone**: Shallow, sunlit near-shore waters where light penetrates to the bottom; supports rooted aquatic macrophytes (water lilies, reeds).
- **Limnetic Zone**: Open, well-lit surface waters away from the shoreline; prime habitat for photosynthetic phytoplankton, zooplankton, and nekton.
- **Profundal Zone**: Deep aphotic open-water layer below the limnetic zone; lacks light for photosynthesis, dominated by heterotrophic decomposers.
- **Benthic Zone**: Bottom floor and sediment layer across the entire body of water inhabited by benthos organisms.
- **Nitrogen-Fixing Symbiosis**:
- **Rhizobium** genus bacteria establish a symbiotic relationship within the root nodules of **Leguminous plants / Pulses** (Fabaceae family, e.g., gram, pea, arhar, lentil, soybean). They convert atmospheric dinitrogen (N_2) into biologically accessible ammonium (NH_4^+) and nitrates, naturally replenishing soil nitrogen fertility without chemical fertilizers.

PYQ — UPPCS Prelims 2025, Q22

With reference to 'Carrying Capacity of Environment', which of the following statements is/are correct?

1. The rate of resource extraction should remain above the rate of resource generation.
2. Generation of waste should remain within the absorption capacity of environment.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

Show answer

Ans: A — Only statement 2. Extraction must stay **below** regeneration, not above.

PYQ — UPPCS Prelims 2025, Q51

Assertion (A): Sustainable development should take place without harming the environment and without compromising the needs of future generations.

Reason (R): Agenda 21 was signed by world leaders in 1995.

A. Both true; R not explanation

B. A false, R true

C. A true, R false

D. Both true; R explains A

Show answer

Ans: C — A is true (Brundtland definition). R is false — Agenda 21 = **Rio 1992**.